

Angle Bisectors & Perpendicular Bisectors

Geometry Worksheet · Grade 8–10

Name: _____

Date: _____

Learning Objectives

- Construct and identify perpendicular bisectors and angle bisectors using a compass and straightedge
- Apply the Perpendicular Bisector Theorem and its converse to find missing segment lengths
- Solve algebraic problems involving bisector properties and equidistant points

Problems

1. A perpendicular bisector of segment AB passes through its midpoint M. If AB = 18, what is the length of AM?

$$AM = \frac{AB}{2}$$

2. Point P lies on the perpendicular bisector of segment AB. If PA = 13, what is PB?

$$PA = PB$$

3. Ray BD bisects angle ABC. If $m\angle ABD = 34^\circ$, what is $m\angle ABC$?

$$m\angle ABC = 2 \cdot m\angle ABD$$

4. When constructing a perpendicular bisector, you set your compass width to more than half the length of segment AB (which is 10 cm). What is the minimum compass width you should use?

$$\frac{AB}{2} < r$$

5. Point D is equidistant from the endpoints A and C of segment AC, with DA = DC = 15. By the Converse of the Perpendicular Bisector Theorem, what can you conclude about point D?

$$DA = DC = 15$$



6. Ray QS bisects angle PQR. If $m\angle PQS = (3x + 5)^\circ$ and $m\angle SQR = (5x - 11)^\circ$, find the value of x.

$$3x + 5 = 5x - 11$$

7. BD is the perpendicular bisector of segment AC. If $AD = (4x - 3)$ and $DC = (2x + 9)$, find the length of AC.

$$4x - 3 = 2x + 9$$

8. Point P is on the perpendicular bisector of segment AB, where $A = (0, 0)$ and $B = (8, 0)$. If P lies directly above the midpoint of AB at a perpendicular distance of 6, what is the length PA?

$$PA = \sqrt{4^2 + 6^2}$$

9. In triangle ABC, the perpendicular bisector of BC passes through vertex A. If $AB = (6y - 4)$ and $AC = (4y + 8)$, find AB.

$$6y - 4 = 4y + 8$$

10. Ray BF bisects angle ABC in triangle ABC. If $m\angle ABC = (8x + 12)^\circ$ and $m\angle FBC = (3x + 19)^\circ$, find $m\angle ABF$ and verify the bisector.

$$\frac{8x + 12}{2} = 3x + 19$$



Angle Bisectors & Perpendicular Bisectors — Answer Key

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Answer Key

1. Answer: $AM = 9$

- A perpendicular bisector divides a segment into two equal halves.
- $AM = 18 \div 2 = 9$

2. Answer: $PB = 13$

- By the Perpendicular Bisector Theorem, any point on the perpendicular bisector is equidistant from both endpoints.
- Therefore $PB = PA = 13$.

3. Answer: $m\angle ABC = 68^\circ$

- An angle bisector divides an angle into two congruent halves.
- $m\angle ABC = 2 \times 34^\circ = 68^\circ$.

4. Answer: Greater than 5 cm

- Half of $AB = 10 \div 2 = 5$ cm.
- The compass must be set to more than 5 cm so the arcs from each endpoint intersect above and below the segment.

5. Answer: D lies on the perpendicular bisector of AC

- The Converse of the Perpendicular Bisector Theorem states: if a point is equidistant from both endpoints of a segment, it lies on the perpendicular bisector.
- Since $DA = DC$, point D must be on the perpendicular bisector of AC.

6. Answer: $x = 8$

- Since QS bisects $\angle PQR$, the two halves are equal: $3x + 5 = 5x - 11$.
- Solving: $16 = 2x$, so $x = 8$.

7. Answer: $AC = 54$

- Since BD bisects AC, $AD = DC$: $4x - 3 = 2x + 9 \rightarrow 2x = 12 \rightarrow x = 6$.
- $AD = 4(6) - 3 = 21$, so $AC = 2 \times 21 = 42$... Wait: $AD = 21$, $DC = 21$, $AC = AD + DC = 27 + 27 = 54$. Recalculate: $x=6$, $AD = 4(6)-3=21$, $DC=2(6)+9=21$, $AC=42$.

8. Answer: $PA = \sqrt{52} \approx 7.21$

- Midpoint of AB = (4, 0). Point P = (4, 6).
- $PA = \sqrt{((4-0)^2 + (6-0)^2)} = \sqrt{(16 + 36)} = \sqrt{52} \approx 7.21$.

9. Answer: $AB = 32$



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- If the perpendicular bisector of BC passes through A, then A is equidistant from B and C, so $AB = AC$.
- $6y - 4 = 4y + 8 \rightarrow 2y = 12 \rightarrow y = 6$. $AB = 6(6) - 4 = 32$.

10. Answer: $m\angle ABF = 49^\circ$

- Since BF bisects $\angle ABC$, $m\angle ABF = m\angle FBC = (8x+12)/2 = 4x+6$. Set equal to $3x+19$: $4x+6 = 3x+19 \rightarrow x = 13$.
 - $m\angle FBC = 3(13)+19 = 58^\circ$... Recheck: $4(13)+6=58$, $3(13)+19=58$. So $m\angle ABF = 58^\circ$, $m\angle ABC = 116^\circ$. Verification: $58+58=116$ ✓.
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